

Episodic Precipitation: Supplementary Information

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I. OVERVIEW

In Sect. II we describe our setup and data analysis, and we described the movies that are also provided in the supplementary information. Phase diagrams of the considered materials are provided in Sect. III, and in Sect. IV we document how the protocols for the temperature ramp are determined. Formal details on the derivation of the oscillation period Δt are filled in in Sect. V. Finally, in Sect. VI in this section we show the unprocessed data on the dependence of Δt on ξ and provide the temperature dependence of all material constants that are used to determine the master plots, Fig. P3 and P4. Here and in the following we refer to figures and equations of the article by preceding their labels by the letter ‘P’, like paper.

II. EXPERIMENTAL METHOD

A picture of the experimental setup is provided in Fig. 1.a. The sample cell (1), a 3 ml fluorescence cell 117.100F-QS made by Hellma GmbH, is illuminated by a KL 2500 LCD Schott cold light source (2) such that dark-field images can be taken with a BM-500CL monochrome progressive scan CCD camera (3). It takes $2,058 \times 2,448$ pixel images with a frame rate between 0.1 and 3 Hz depending on the ramp rate ξ . The sample temperature is controlled by immersion into a water bath that follows a temperature protocol imposed by a computer-controlled thermostat (4): an immersion cooler Haake EK20 is cooling with constant power, and a Huber CC-E immersion thermostat is heating the water bath to the preset temperature. Additionally, the temperature of the water near the sample is measured with a PT100 temperature sensor. The temperature is controlled with an accuracy of 15 mK. Homogenisation for repeated runs is provided by a magnetic stirring unit (5).

The inset (1a) shows a magnification of the sample cell. The camera captures the turbidity of the full cell, providing 8 bit turbidity data as shown in Fig. 1.c. Averaging this data in horizontal direction and plotting the resulting scans of the turbidity height profiles, provides the space-time plot Fig. 1.b. For visual inspection the contrast in these pictures is conveniently enhanced by a representation in false colours, Fig. P1. In the supplementary online information we also provide movies showing the black-and-white turbidity data of the camera together with a plot of the time evolution of the temperature, and an animation illustrating the construction of the false-colour plot of the turbidity shown in Fig. P1.g.

The data shows the phase separation of binary mixtures subjected to a ramp rate, ξ , describing the linear growth of the volume fraction of droplets. Due to the ramp one observes

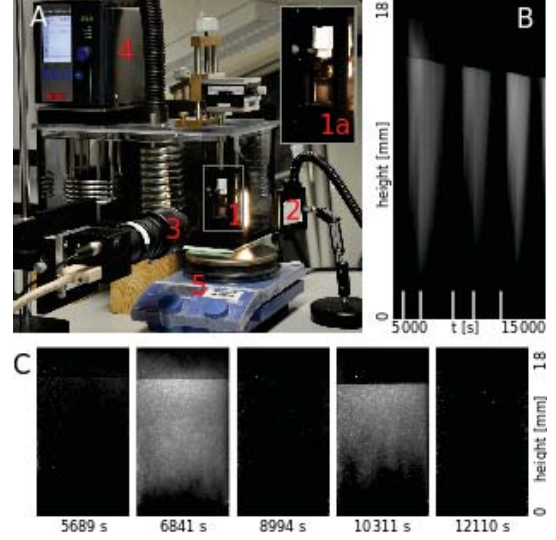


FIG. 1. **Setup and raw data obtained from the CCD camera.** a. Photo of the experimental setup. Its components, (1)–(4) are described in the main text. b. Space-time plot based on the turbidity data provided by the camera. c. Representative snap shots of a sample containing a mass fraction of 0.2751 isobutoxyethanol in water, subjected to a ramp rate $\xi = 2.5 \times 10^{-5} \text{ s}^{-1}$. The times where the snap shots were taken are indicated below the respective photos, and by white bars in panel b.

repeated waves of precipitation. They go along with marked oscillatory changes in the droplet size distribution,¹ and in the turbidity of the samples.² The main panels of Fig. 2 show representative traces of the turbidity of the samples when heated with different constant ξ . Details on the experimental setup and on the data analysis to extract the oscillation period from the space-time plots were provided in Ref. 2

III. PHASE DIAGRAMS

We investigate the evolution of the domain structure (droplet formation, growth and sedimentation) in mixtures where the temperature is changed gradually in order to continuously drive the phase separation of a fluid binary mixture. Two types of mixtures are considered:

- i. Methanol/hexane (M + H):** These mixtures are one of the classical model systems of binary phase separation.^{3–7} The two liquids are fully miscible for temperature above $T_c = 34.45^\circ\text{C}$. The concentrations of the coexist-

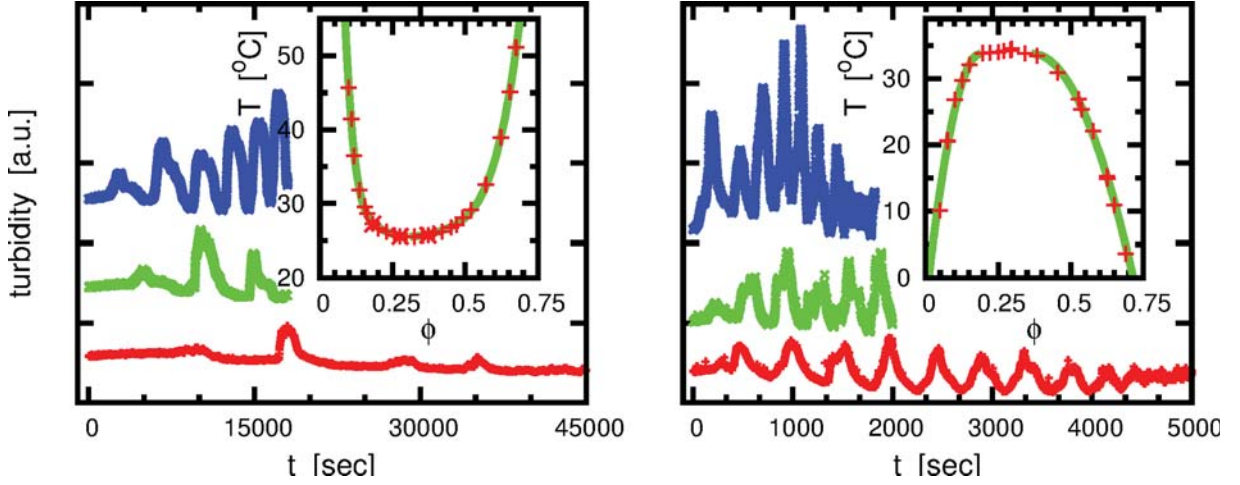


FIG. 2. **Phase diagrams and time traces of the turbidity.** The insets show the phase diagrams of IBE/water (left), and methanol/hexane (right), and the respective main panels show oscillations of the turbidity in the lower layer for different ramp rates ξ : from top to bottom (left) $2.5 \times 10^{-5} \text{s}^{-1}$, $1.25 \times 10^{-5} \text{s}^{-1}$, $5 \times 10^{-6} \text{s}^{-1}$ for IBE/water, and (right) $4 \times 10^{-4} \text{s}^{-1}$, $2 \times 10^{-4} \text{s}^{-1}$, $1 \times 10^{-4} \text{s}^{-1}$ for methanol/hexane. The data points of the different turbidity signals are shifted for better visibility.

ing phases that are formed for lower temperatures are shown in the phase diagram in Fig. 2(right). See Ref. 5 for a more detailed description.

ii. Isobutoxyethanol/water (IBE + W): Mixtures of water and butoxyethanol have become popular as an experimentally-friendly system that separates upon heating⁸. For our present purposes IBE and water (Fig. 2(left)) is even preferential since the critical point of this mixtures, $T_c = 25.61^\circ\text{C}$, lies more than 10°C below the one of the butoxyethanol mixture. This further enhances the range of experimentally accessible temperatures (they must always lie well below the boiling point of water). See Ref. 1 for a more detailed description.

For the fit of the coexistence curve we follow the procedure of Ref. 9. To first order they approximate the left and right branch of the coexistence curve by

$$\Phi_{r/l} = \Phi_c \pm B \theta^\beta + D \theta^{2\beta} \quad (1)$$

with the reduced temperature $\theta = |1 - T/T_c|$, the critical point being at temperature T_c with composition Φ_c and the universal scaling exponent $\beta = 0.325$. This provides a good fit, the solid green line the inset of Fig. 2(right), for the M + H mixture, with fit parameters listed in Tab. I. On the other hand¹⁰ for IBE + W the exponent $\beta = 0.325$ only applies for $\theta < 10^{-3}$, which is too small for our purposes. Even correction terms based on the Wegner expansion do not help.¹⁰ To have a simple set of parameters we therefore choose $\beta = 0.25$, which admits a faithful description based on three free non trivial parameters (see Fig. 2(left) and Tab. I).

The temperature ramps in our experiments amount to increasing temperature for IBE/water, and decreasing temperature for methanol/hexane. For simplicity we denote this as

	IBE + W	M + H
β	0.25	0.325
T_c [°C]	25.61 ± 0.12	34.45 ± 0.14
Φ_c	0.3093 ± 0.0032	0.31368 ± 0.00077
B	0.2958 ± 0.0013	0.358 ± 0.001
D	0.076 ± 0.004	0.0794 ± 0.0012

TABLE I. Fit parameters of the coexistence curve for IBE + W and M + H.

heating, and understand that the temperature ramp rate is negative for the latter mixture.

The evolution can most conveniently be described by focusing on a region in one of the macroscopic phases. Its average concentration changes due to sedimentation of large droplets. However, immediately after a precipitation event, the bulk and the remaining small droplets are in equilibrium at points on the coexistence curve with composition Φ_b for the bulk phase, and Φ_d for the remaining droplets in the solution. The phases occupy the volumes V_b and V_d , respectively.

As the mixture is further heated, the equilibrium concentrations of the coexisting phases changes in response to the broadening of the miscibility gap, *i.e.* the region bounded by the coexistence curve. A temperature difference δT causes a change in the equilibrium composition by $\delta \Phi_b < 0$ and $\delta \Phi_d > 0$. It gives rise to a concentration current across the interface of the droplets which in turn leads to a growth of the droplets. In the following section we review how the temperature protocol of the experiments was chosen in order to fix the growth of the droplet volume fraction.

IV. CALCULATING THE RAMP RATE ξ

A straightforward derivation of the ramp rate starts from the the average composition

$$\phi = v_d \Phi_d + (1 - v_d) \Phi_b \quad (2)$$

of a small volume of a mixture, where droplets of composition Φ_d occupy a volume fraction v_d in a background phase of composition Φ_b . By definition, this average composition is preserved when the droplets start growing in response to a change of temperature. On the other hand droplet growth is accompanied by a change of the composition of the phases,

$$0 = \dot{\phi} = v_d \dot{\Phi}_d + \Phi_d \dot{v}_d + (1 - v_d) \dot{\Phi}_b - \Phi_b \dot{v}_d \quad (3)$$

Introducing the notations

$$\zeta = \Phi_0^{-1} \frac{d\bar{\Phi}}{dt} \quad (4a)$$

$$\xi = \Phi_0^{-1} \frac{d\Phi_0}{dt} \quad (4b)$$

$$\varphi = \frac{\phi - \bar{\Phi}}{\Phi_0} \quad (4c)$$

$$\text{where } \bar{\Phi} = \frac{1}{2} (\Phi_b + \Phi_d), \quad (4d)$$

$$\Phi_0 = \frac{1}{2} (\Phi_b - \Phi_d), \quad (4e)$$

substituting the resulting expressions for $\dot{\Phi}_d$ and $\dot{\Phi}_b$ into Eq. (3), and solving for $\dot{v}_d$ one obtains after some straightforward algebra

$$\dot{v}_d = \frac{1}{2} [\zeta + \xi \varphi]. \quad (5)$$

Here, φ is the reduced average concentration. It takes the value one when $\phi = \Phi_b$ and smaller values for compositions inside the miscibility gap.

Assuming local equilibrium one can characterise the local bulk concentration by the space dependent field $\varphi(x, t)$. Its time evolution obeys a diffusion equation¹¹ with a source strength of $2\dot{v}_d$. According to the above consideration this source term then gives rise to a corresponding growth of the equilibrium droplet volume fraction. From the point of view of the transport equations, the magnitude of the source strength $\dot{v}_d$ appears therefore as the relevant parameter characterising how strongly the mixture is driven away from equilibrium. With this motivation we consider here temperature protocols that correspond to fixed values of $\dot{v}_d$.

In Eq. (4) it is understood that Φ_0 and $\bar{\Phi}$ are functions of $T(t)$ due to their dependence of Φ_d and Φ_b , *i.e.* on the borders of the two phase region of the phase diagram. In general these two functions have a different temperature dependence such that it is not clear a priori that $\dot{v}_d$ can be fixed to a constant value by choosing an appropriate form of the temperature ramp $T(t)$. Indeed it is necessary to choose different

temperature protocols for the two phases — *i.e.* for instance different temperature ramps for looking at methanol droplets in hexane than for hexane droplets in methanol. The optimal protocol can be found by rearranging Eq. (5) to take the form

$$\dot{v}_d = \frac{1}{2\Phi_0(T)} \frac{d\Phi_b}{dt} - \frac{v_d}{\Phi_0} \frac{d\Phi_0}{dt} \approx \frac{1}{2\Phi_0(T)} \frac{d\Phi_b}{dT} \frac{dT}{dt}, \quad (6)$$

where the approximation in the final step is based on the fact that the volume fraction of droplets v_d is always small in our experiments. According to this expression the ramp rate $\dot{v}_d$ for droplets in the upper and lower phase of our samples is found by appropriately assigning the indices b and d to the respective branches of the phase diagram. Subsequently, the temperature protocol of the ramp is obtained by integrating

$$\frac{dT}{dt} = 2\dot{v}_d \Phi_0(T) \left(\frac{d\Phi_d}{dT} \right)^{-1} \quad (7)$$

In practice there is only a small difference between $\dot{v}_d$ and ξ since $\zeta \ll \xi$ for the phase diagrams under consideration, and since φ will always be very close to one. In view of this we only distinguish between $\dot{v}_d$ and ξ for the sake of calculating of the temperature protocol. This avoids systematic errors in the numerical integration of Eq. (7). For the purpose of the further presentation of the data, we specify the ramp rate in terms of ξ in order to be consistent with the pertinent literature.^{1,2,11–13}

V. CALCULATING THE PERIOD Δt

In this section we provide a formal derivation of the parameter dependence of the oscillation period Δt . In Sect. VI we provide all information constants required for a quantitative comparison of the prediction with the experimental data. The derivation will be based on the equation for the droplet growth

$$\dot{r} = \frac{\xi}{4\pi n} \frac{1}{r^2} + \frac{\epsilon \kappa \xi t}{4} r^2 \quad (P3)$$

which must be solved for the initial conditions that $r(t=0) = 0$. The equation involves two contributions. At early times the droplets are small, and $\xi/(4\pi n r^2)$ dominates. It accounts for diffusive growth shared by particles that give rise to a droplet density n . Later on, the largest particles start to feel the buoyancy. They sediment and collect small particles which gives rise to a contribution $\epsilon \kappa \xi t r^2/4$. In the latter term the parameter ϵ is the collision efficiency, and κ accounts for the parameter dependence of the terminal velocity, u , of a sphere moving slowly through a viscous medium. This constant velocity results from balancing the droplet buoyancy with the Stokes drag,

$$u = \kappa r^2 \quad \text{with} \quad \kappa = \frac{2g \Delta \rho}{9\nu \rho}, \quad (8)$$

where ν is the kinematic viscosity, ρ is the density of the majority phase, g is the gravitational acceleration, and $\Delta \rho$ the density difference between the phases.

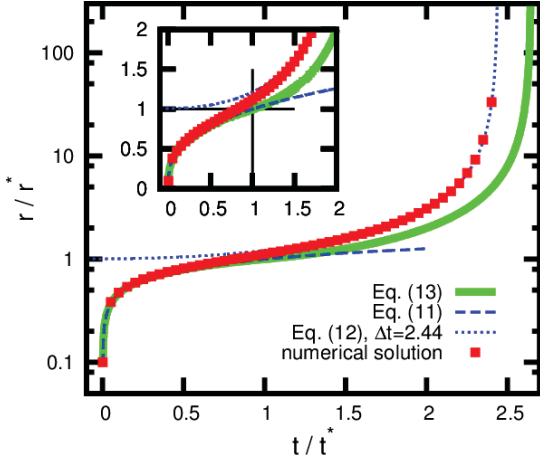


FIG. 3. **Time evolution of the droplet radius.** The red boxes show the numerical solution of Eq. (10), and the green thick solid line the analytical approximations, Eqs. (13). The dashed and dotted blue lines show the approximations for small and large droplets, Eqs. (11) and (12), respectively. The inset shows the same data in a Cartesian coordinate system zooming into the cross-over region at the bottom of the bottleneck.

Henceforth, we measure time in units of

$$t^* \sim \left(\frac{n}{\epsilon^3 \kappa^3 \xi^4} \right)^{1/7}. \quad (9a)$$

and length in units of

$$r^* = r(t^*) \simeq [n \epsilon \kappa t]^{-1/4}, \quad (9b)$$

and, for conciseness of the notation, we denote the resulting dimensionless units still as (r, t) . In terms of these dimensionless variables Eq. (P3) takes the form

$$\dot{r} = \frac{1}{3r^2} + \frac{tr^2}{3} \quad (10)$$

The growth velocity $\dot{r}(r, t)$ takes its minimum at $(r, t) = (1, 1)$. Consequently, t^* denotes the time required for droplets to grow from zero radius to the bottleneck radius r^* where the droplet growth speed, $\dot{r}$, is minimal.

As long as buoyancy does not yet affect the motion of the largest droplets in the mixture, they grow diffusively by collecting supersaturation. In leading order for small droplets one can then neglect the growth contribution $r^2 t$. For the initial droplet size $r(t=0) = 0$ this entails

$$\dot{r} \simeq \frac{1}{3r^2} \Rightarrow r(t) \simeq t^{1/3} \quad \text{for } r \ll r^*. \quad (11)$$

As shown by the dashed blue line in Fig. 3 this approximation provides a good estimate for values $t < t^*/2$.

Similarly, for large droplets the contribution r^{-2} to the growth is sub-dominant such that in leading order

$$\dot{r} \simeq \frac{r^2 t}{3} \Rightarrow r(t) \simeq \frac{6}{\Delta t^2 - t^2} \quad \text{for } r \gg r^*. \quad (12)$$

Hence, the growth features a finite-time singularity when t approaches Δt . At the latest at this late time, the droplets will rapidly fall out of the measurement window, such that the mixture is reset to its initial state $r = 0$.

For the choice $\Delta t = 2.44$ Eq. (12) provides a very good description of the numerical data for $t \gtrsim t^*$. On the other hand, for this choice of Δt the expression (12) can not be matched continuously to Eqs. (11) because these expressions produce smaller values for $r(t)$ for all t . Rather, a continuous interpolation from Eq. (11) to (12) requires to choose $t = t^*$ and $r = r^*$,

$$r(t) \simeq \begin{cases} t^{1/3} & \text{for } t \leq 1 \\ 6(7-t^2)^{-1} & \text{for } t \geq 1 \end{cases} \quad (13)$$

The resulting first order estimate for $r(t)$ is shown by the solid line in Fig. 3. This expression slightly over-estimates the time Δt required to reach the finite time singularity.

This analysis of Δt shows that the time scale for the growth from vanishingly small to very large droplets is a small multiple of the time, t^* , required to grow to the bottleneck size, r^* . In particular, t^* provides the parameter dependence of the time Δt between subsequent precipitation events, because according to Eq. (10) it is the only time scale involved in the description of the growth from tiny to huge droplets,

$$\Delta t \sim t^* \sim \left(\frac{n}{\kappa^3 \xi^4} \right)^{1/7}. \quad (P5)$$

Sugimoto¹⁴ pointed out in Eqs. (1)–(3) of his paper that in the presence of a molar feed rate of monomers, Q , the number density, n , of droplets obeys

$$n \sim \frac{Q RT}{\gamma V_m C_\infty D} \quad (14)$$

where R is the molar gas constant, T the temperature, γ the interfacial tension, V_m the molar volume, C_∞ is the equilibrium composition of the droplet phase in units of mol/m³, and D is the pertinent diffusion coefficient. This finding can be rewritten in a more compact form by introducing the Kelvin length^{15,16}

$$\sigma = \frac{2\gamma V_m^2 C_\infty}{RT}, \quad (15)$$

and $Q = \xi/V_m$, into Eq. (14),

$$n \sim \frac{\xi}{\sigma D}. \quad (16)$$

Inserting Eq. (16) into Eq. (P5) provides

$$\Delta t \sim t^* \sim (D\sigma\kappa^3)^{-1/7} \xi^{-3/7}, \quad (P6)$$

A first hint that Eq. (P6) is a faithful prediction can be obtained by observing that the parts of a period where we observe high and low turbidity in figure 2 are of comparable extent. This is consistent with the theoretical prediction that the singularity arises at $\Delta t = 2.44 t^*$. For the *quantitative* test, that is given in the paper, one must take into account that the first factor of this scaling law involves material constants with a non-trivial temperature dependence.

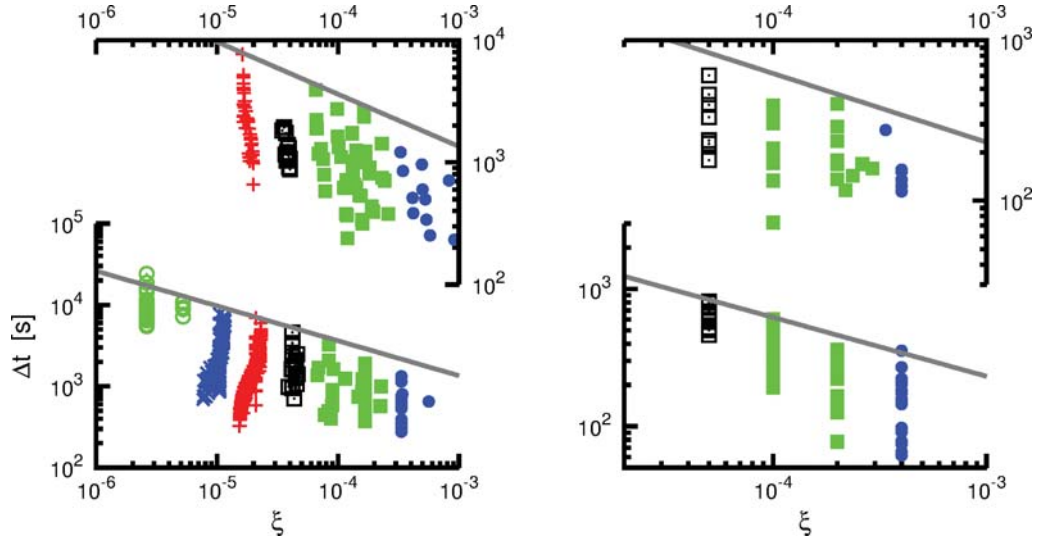


FIG. 4. **The oscillation period, Δt , plotted vs. the ramp rate ξ .** The four panels show Δt for IBE-rich bubbles in a water-rich continuous phase (bottom left), water-rich droplets in an IBE-rich continuous phase (top left), hexane-rich bubbles in a methanol-rich continuous phase (bottom right), and methanol-rich droplets in a hexane-rich continuous phase (top right), respectively. The different symbols denote measurements for different ranges of ramp rates, ξ . The plots show the same data and use the same symbols as the master plots Fig. P4. In all cases the height of the layer was $h \approx 1$ cm. Measurements with $h = 0.5$ cm and $h = 0.25$ cm for the lower phase showed that Δt hardly changes as h is reduced. The data points for the IBE/water mixture (left) are obtained by particle tracking (*cf.* Ref. 1 for experimental details), and those for methanol/hexane (right) refer to subsequent minima of turbidity measurements as shown in Fig. 2. We verified that both methods provide the same results. However, the data obtained from droplet tracking tend to be more accurate. The grey lines are guides to the eye that indicate the slope of a power law, $\Delta t \sim \xi^{-3/7}$. The data is also provided as an ASCII file in the supplementary information.

VI. MATERIAL CONSTANTS

Fig. 4 shows the period, Δt , for various measurements at different ramp rates, ξ . Different data points for a given ramp rate are due to the drift of Δt when the pertinent material constants, D , σ , and $\kappa(\Delta\rho, \rho, \nu)$ change upon moving further away from the critical point. In the following we provide the temperature dependencies of these data, which in turn translate to a time dependence when inverting the protocol $T(t)$ of the temperature ramp. Upon doing so we denote the mass fraction as ϕ^m and the molar fraction as ϕ^n , respectively. We cite the data here as they were provided in the original literature (even when we have doubts that they are accurate to six significant digits for our samples).

A. IBE and water

The theoretical curves in Fig. P4 use data on material parameters from a variety of sources.^{17–21} The index $i \in \{\text{IBE}, \text{W}\}$ will be used to refer to material properties of IBE and water, respectively, and in accordance with the phase diagram Fig. 2(left) the concentration are always given in terms of $\phi = \phi_{\text{IBE}}$.

a. *Density (based on Ref. 19)* The densities of the phases are determined by the composition, thermal expansion

and molar excess volume,

$$\rho(\phi^m, T) = \left[\frac{\phi^m}{\rho_{\text{IBE}}} + \frac{1 - \phi^m}{\rho_{\text{W}}} + \left(\frac{\phi^m}{M_{\text{IBE}}} + \frac{1 - \phi^m}{M_{\text{W}}} \right) V_{\text{E}}^n \right]^{-1}. \quad (17)$$

where $\rho_i = \rho_i(T)$ are the (temperature-dependent) densities of the pure substances, M_i their molar masses, and $V_{\text{E}}^n = V_{\text{E}}^n(\phi^n)$ is the molar excess volume.

The molar masses, M_i are 18.01528 g/mol for water²² and 118.17416 g/mol for IBE,²³ respectively.

The temperature dependence of the density, $\rho_i(T)$, of the pure substances is linearly approximated around $T_0 = 25^\circ\text{C}$,

$$\rho_i(T) = \rho_i(T_0) - \alpha_i (T - T_0) \quad (18)$$

with fit parameters for ρ_i and α_i given in table II.

Moreover, the molar excess volume is fitted according to

	$\rho_i(T_0)$ [g cm ⁻³]	α_i [g cm ⁻³ K ⁻¹]
water	0.997043	0.2571×10^{-3}
IBE	0.886255	0.968×10^{-3}

TABLE II. Densities and thermal expansion coefficients for water and IBE¹⁹.

	A [kg m ⁻¹ s ⁻¹]	B	C [(°C) ⁻¹]	D [°C]
water	1.002×10^{-3}	1.3272	0.001053	105
IBE	3.4×10^{-3}	1.7	0.001	110

TABLE III. Data of Ref. 24 for the fit coefficients for the viscosity of water and IBE, defined by equation Eq. (20). In both cases $T_0 = 20^\circ\text{C}$ is used as reference temperature.

Ref. 19:

$$V_E^n(\phi^n) = \frac{\phi^n (1 - \phi^n)}{1 - G \tilde{\phi}} \left(A_1 + A_2 \tilde{\phi} + A_3 \tilde{\phi}^2 \right) \quad (19)$$

$$\text{with } \tilde{\phi} = 1 - 2\phi^n$$

$$\text{and } G = 0.975,$$

$$A_1 = -3.079 \text{ cm}^3/\text{mol},$$

$$A_2 = 1.801 \text{ cm}^3/\text{mol},$$

$$A_3 = 0.839 \text{ cm}^3/\text{mol}.$$

A slight temperature dependence of these fit parameters was

$$\eta(\phi^m, T = 25^\circ\text{C}) = -40.66 (\phi^m)^5 + 103.44 (\phi^m)^4 - 100.32 (\phi^m)^3 + 39.35 (\phi^m)^2 + 0.17 \phi^m + 0.91. \quad (21a)$$

$$= w_\eta(\phi^m) \eta_{\text{IBE}}(T = 25^\circ\text{C}) + [1 - w_\eta(\phi^m)] \eta_{\text{W}}(T = 25^\circ\text{C}). \quad (21b)$$

where the latter equation *defines* the dimensionless, non-linear weight function w_η that expresses $\eta(\phi^m, T = 25^\circ\text{C})$ as a function of the viscosities of the pure substances $\eta_{\text{IBE}}(T = 25^\circ\text{C})$ and $\eta_{\text{W}}(T = 25^\circ\text{C})$, respectively.

Assuming that this weight function is not varying substantially in the temperature range of our measurements, we can use the interpolation (21b) to determine the viscosity of the mixture also at other temperatures. After all, the temperature dependence $\eta_{\text{IBE}}(T)$ and $\eta_{\text{W}}(T)$ were provided by Eq. (20) with coefficients in table III. To check the strong assumption entering this interpolation,

$$\eta(\phi^m, T) = w_\eta(\phi^m) \eta_{\text{IBE}}(T) + [1 - w_\eta(\phi^m)] \eta_{\text{W}}(T). \quad (22)$$

we measured the viscosity of the two coexisting phases at $T = 40^\circ\text{C}$. For both phases the prediction of Eq. (22) was accurate to within 2%. This is sufficient for our purposes.

c. Diffusion coefficient (based on Ref. 17) The renormalisation group theory predicts that the diffusion coefficient vanishes when the critical point is approached. On the other hand, the renormalisation group theory is precise only in the vicinity of the critical point, and its application to interdiffusion coefficients has been a source of controversy.^{25,26} For this reason we choose to rely upon interpolations of experimental data. We do not expect that our data follow the critical exponents because the temperatures in our experiments lie outside the critical region. Hence, we fitted the data of Ref. 17 with

reported in Ref. 19. However, it is so small that we need not take it into account here.

To get the dependence of the density difference on the reduced temperature the dependence $\phi(\theta)$ (coexistence curve) has to be inserted into Eq. (17).

b. Viscosity (own measurements augmented by data of Refs. 24 and 20) We first provide the data of the pure phases, and then obtain the viscosity of the mixture by appropriate interpolation.

Following Ref. 24 we describe the temperature dependence of the pure substances by

$$\eta_i(T) = A_i 10^{\frac{B_i (T_0 - T) - C_i (T_0 - T)^2}{T + D_i}}. \quad (20)$$

In table III we provide the values for pure water provided in Ref. 24, and parameters of a fit for IBE whose viscosity we determined with an Ubbelohde viscometer type 537 10/I made by Schott.

To interpolate the viscosities for a mixed phase of given mass fraction ϕ^m we use the composition-dependent viscosities at 25°C for a homogeneous mixture in the single-phase regime.²⁰ The data is fitted with a fifth order polynomial

the following expression:

$$D_i(\theta) = D_c + \delta_i \theta \quad (23)$$

with $D_c = 6.4 \times 10^{-12} \text{ m}^2/\text{s}$, $\delta_{\text{IBE}} = 2.1 \times 10^{-9} \text{ m}^2/\text{s}$ and $\delta_{\text{W}} = 4.1 \times 10^{-9} \text{ m}^2/\text{s}$.

d. Interfacial tension (based on Ref. 18) The interfacial tension vanishes at the critical temperature, and its dependence at higher temperatures can be represented by a power law

$$\gamma(\theta) = \gamma_0 \theta^{\alpha_\gamma}. \quad (24)$$

where a fit to the data of Ref. 18 yields $\gamma_0 = 7.3 \times 10^{-4} \text{ N/m}$ and $\alpha_\gamma = 1.2$.

e. Molar volume (based on Ref. 21) According to Ref. 21 the molar volume V^n can be approximated by

$$V^n = \phi^n V_{\text{IBE}}^n + (1 - \phi^n) V_{\text{W}}^n \quad (25)$$

with $V_{\text{IBE}}^n = 124 \text{ cm}^3/\text{mol}$ and $V_{\text{W}}^n = 15.98 \text{ cm}^3/\text{mol}$.

B. Methanol/hexane mixtures

In this subsection the index $i \in \{M, H\}$ denotes material constants of methanol/hexane, respectively, and concentrations refer to methanol, $\phi = \phi_M$.

	a_0 [g cm ⁻³]	a_1 [g cm ⁻³ K ⁻¹]	a_2 [g cm ⁻³ K ⁻²]
methanol	1.382	-3.135×10^{-3}	3.813×10^{-6}
hexane	0.6839	6.989×10^{-4}	-2.656×10^{-6}

TABLE IV. Coefficients of the density.⁵

f. Density (based on Refs. 5 and 27) The densities are again calculated according to Eq. (17). In this case the molar mass is 32.04186 g/mol for methanol²⁸ and 86.17536 g/mol for hexane.²⁹ The temperature dependence of the pure substances amounts to⁵

$$\rho_i(T) = a_0 + a_1 T + a_2 T^2 \quad (26)$$

with coefficients given in table IV, and the excess volume can be expressed²⁷ as

$$V_E^n(\phi^n) = \phi^n (1 - \phi^n) [B_0 + B_1 \tilde{\phi} + B_2 \tilde{\phi}^2] \quad (27a)$$

$$\text{with } \tilde{\phi} = 1 - 2\phi^n$$

$$\begin{aligned} \text{and } B_0 &= 2.0741 \text{ cm}^3/\text{mol}, \\ B_1 &= 0.3195 \text{ cm}^3/\text{mol}, \\ B_2 &= 1.7733 \text{ cm}^3/\text{mol}. \end{aligned}$$

g. Viscosity (based on data of Refs. 30, 31 and 27) We first provide the data of the pure phases, and then obtain the viscosity of the mixture by appropriate interpolation.

The viscosity of pure methanol³⁰ is

$$\eta_M(T) = A \exp(B/T) \quad (28)$$

with $A = 8.203 \times 10^{-6} \text{ Pa s}$ and $B = 1251.4 \text{ K}$.

For hexane our analysis is based on the kinematic viscosity ν_H provided in Ref. 31

$$\nu_H(T) = \nu' \left(\frac{T}{T'} \right)^n \exp \left(\frac{B(T' - T)}{(T' - T_0)(T - T_0)} \right) \quad (29)$$

with $n = -2.24057$, $B = 4.78496 \text{ K}$ and $T_0 = 222.468 \text{ K}$, reference viscosity $\nu' = 0.4604 \times 10^{-10} \text{ m}^2/\text{s}$, and reference temperature $T' = 296.267 \text{ K}$. Together with the density of hexane, which is provided in Eq. (26), this provides the dynamic viscosity $\eta_H = \rho_H \nu_H$.

The viscosity of the mixture is obtained by interpolating using the excess viscosity provided in Ref. 27

$$\begin{aligned} \eta(\phi^n, T) &= \phi^n \eta_M(T) + (1 - \phi^n) \eta_H(T) \\ &\quad + \phi^n (1 - \phi^n) [B_0 + B_1(1 - 2\phi^n)] \quad (30) \end{aligned}$$

$$\text{with } B_0 = -1.83 \times 10^{-4} \text{ kg m}^{-1} \text{ s}^{-1}$$

$$B_1 = 0.91 \times 10^{-4} \text{ kg m}^{-1} \text{ s}^{-1}.$$

h. Diffusion coefficient (based on Ref. 32) The dependence of the diffusion coefficient $D(\phi^n, \theta)$ on the concentration ϕ^n of the mixture and on the reduced temperature θ can be approximated by³²

$$\begin{aligned} D(\phi^n, \theta) &= A_0 + A_1 \phi^n + A_2 (\phi^n)^2 + A_3 (\phi^n)^3 + A_4 (\phi^n)^4 \\ &\quad + A_5 \theta^{0.68516} \quad (31) \end{aligned}$$

$$\text{with } A_0 = 3.2457 \times 10^{-9} \text{ m}^2/\text{s},$$

$$A_1 = -1.68497 \times 10^{-8} \text{ m}^2/\text{s},$$

$$A_2 = 3.63103 \times 10^{-8} \text{ m}^2/\text{s},$$

$$A_3 = -4.1949 \times 10^{-8} \text{ m}^2/\text{s},$$

$$A_4 = 2.223 \times 10^{-8} \text{ m}^2/\text{s},$$

$$A_5 = 2.5067 \times 10^{-9} \text{ m}^2/\text{s}. \quad (32)$$

In order to accurately account for the composition dependence of the background contribution these authors resorted to a forth-order polynomial to describe its dependence on ϕ^n . The term $A_5 \theta^{0.68516}$ accounts for the singular contribution to the diffusivity. It vanishes at $T = T_c$ with the appropriate scaling exponent, 0.68516. Similar to the expression (23) the fit for the methanol/hexane mixture hence also involves a non-singular, constant background contribution.

i. Interfacial tension (according to Ref. 5) Data of interfacial tension⁵ can be approximated according to Eq. (24) with $\gamma_0 = 3.631 \times 10^{-2} \text{ N/m}$ and $\alpha_\gamma = 1.65$. This data lies beyond the critical region of $\theta < 10^{-2.5}$ where scaling with a critical exponents is expected.⁵

j. Molar volume (according to Ref. 33) The molar volume is interpolated with Eq. (25) with $V^n = 41.1 \text{ cm}^3/\text{mol}$ for methanol and $V^n = 133.2 \text{ cm}^3/\text{mol}$ for hexane.³³

¹ Lapp, T., Rohloff, M., Vollmer, J., and Hof, B. Particle tracking for polydisperse sedimenting droplets in phase separation. *Exp. Fluids* **52**, 1187–1200 (2012).

² Auernhammer, G. K., Vollmer, D., and Vollmer, J. Oscillatory instabilities in phase separation of binary mixtures: Fixing the thermodynamic driving. *J. Chem. Phys.* **123**, 134511 (2005).

³ Huang, J. S., Goldburg, W., and Bjerkaas, A. W. Study of phase separation in a critical binary liquid mixture: spinodal decomposition. *Phys. Rev. Lett.* **32**, 921–923 (1974).

⁴ Beysens, D., Guenoun, P., and Perrot, F. Phase separation of critical binary fluids under microgravity: Comparison with matched-density conditions. *Phys. Rev. A* **38**, 4173–4185 (1988).

⁵ Abbas, S., Satherley, J., and Penfold, R. The liquid-liquid coexistence curve and the interfacial tension of the methanol-n-hexane system. *J. Chem. Soc. Farad Trans.* **93**, 2083–2089 (1997).

⁶ Iwanowski, I., Sattarow, A., Behrends, R., Mirzaev, S. Z., and Kaatz, U. Dynamic scaling of the critical binary mixture methanol-hexane. *J. Chem. Phys.* **124**, 144505 (2006).

- ⁷ Sam, E. M., Auernhammer, Y. H. G. K., and Vollmer, D. Pattern formation in phase separating binary mixtures. *Phys. Chem. Chem. Phys.* **13**, 13333 – 13340 (2011).
- ⁸ Emmanuel, S. and Berkowitz, B. An experimental analogue for convection and phase separation in hydrothermal systems. *J. Geophys. Res.* **111**, B09103 (2006).
- ⁹ Aizpiri, A. G., Correa, J. A., Rubio, R. G., and Pena, M. D. Coexistence curve of methanol+n-heptane: Range of simple scaling and critical amplitudes. *Phys. Rev. B* **41**, 9003 (1990).
- ¹⁰ Nakata, M., Dobashi, T., Kuwahara, N., and Kaneko, M. Coexistence curve and diameter of the system ethylene glycol monoisobutyl ether + water. *J. Chem. Soc. Farad Trans.* **78**, 1801–1810 (1982).
- ¹¹ Cates, M. E., Vollmer, J., Wagner, A., and Vollmer, D. Phase separation in binary fluid mixtures with continuously ramped temperature. *Phil. Trans. Roy. Soc. (Lond.) Ser. A* **361**, 793–807 (2003).
- ¹² Vollmer, J., Auernhammer, G. K., and Vollmer, D. Minimal model for phase separation under slow cooling. *Phys. Rev. Lett.* **98**, 115701 (2007).
- ¹³ Vollmer, J. Phase separation under ultra-slow cooling: Onset of nucleation. *J. Chem. Phys.* **129**(16), 164502 (2008).
- ¹⁴ Sugimoto, T. The theory of the nucleation of monodisperse particles in open systems and its application to AgBr systems. *J. Coll. Interf. Sc.* **150**(1), 208 – 225 (1992).
- ¹⁵ Lifshitz, E. M. and Pitaevskii, L. P. *Landau and Lifshitz Course of Theoretical Physics, Vol.10 : Physical Kinetics*. Butterworth-Heinemann, Oxford, (1981).
- ¹⁶ Bray, A. J. Theory of phase-ordering kinetics. *Adv. Phys.* **43**, 357 – 459 (1994).
- ¹⁷ Steinhoff, B. and Woermann, D. Slowing down of the kinetics of liquid/liquid phase separation along the binodal curve of a binary liquid mixture with a miscibility gap approaching the critical point. *J. Chem. Phys.* **103**(20), 8985–8992 (1995).
- ¹⁸ Aratono, M., Nakayama, S., Ikeda, N., and Motomura, K. Thermodynamic consideration on the interface formation of water and ethylene glycol isobutyl ether mixture. *Coll. Polymer Sc.* **268**, 877–82 (1990).
- ¹⁹ Doi, H., Tamura, K., and Murakami, S. Thermodynamic properties of aqueous solution of 2-isobutoxyethanol at T= (293.15, 298.15, and 303.15) K, below and above LCST. *J. Chem. Thermodyn.* **32**(6), 729–741 (2000).
- ²⁰ Menzel, K., Mirzaev, S. Z., and Kaatze, U. Crossover behavior in micellar solutions with lower critical demixing point: Broad-band ultrasonic spectrometry of the isobutoxyethanol-water system. *Phys. Rev. E* **68**(1), 011501 (2003).
- ²¹ Douheret, G., Davis, M. I., Reis, J. C. R., Fjellanger, I. J., Vaage, M. B., and Hoiland, H. Aggregative processes in aqueous solutions of isomeric 2-butoxyethanols at 298.15 K. *Phys. Chem. Chem. Phys.* **4**(24), 6034–6042 (2002).
- ²² PubChem compound database, CID=962 (17 July 2013).
- ²³ PubChem compound database, CID=521158 (17 July 2013).
- ²⁴ Weast, R. C., Astle, M. J., and Beyer, W. H., editors. *CRC Handbook of Chemistry and Physics*. CRC Press, Inc. Boca Raton, Florida, 69. edition, (1988-1989).
- ²⁵ Sengers, J. V. Transport properties of fluids near critical points. *Int. J. Thermophys.* **6**, 203 (1985).
- ²⁶ Das, S. K., Fisher, M. E., Sengers, J. V., Horbach, J., and Binder, K. Critical dynamics in a binary fluid: Simulations and finite-size scaling. *Phys. Rev. Lett.* **97**, 025702 (2006).
- ²⁷ Orge, B., Iglesias, M., Rodriguez, A., Canosa, J. M., and Tojo, J. Mixing properties of (methanol, ethanol, or 1-propanol) with (n-pentane, n-hexane, n-heptane and n-octane) at 298.15 K. *Fluid Phase Equilibria* **133**(1-2), 213–227 (1997).
- ²⁸ PubChem compound database, CID=887 (17 July 2013).
- ²⁹ PubChem compound database, CID=8058 (17 July 2013).
- ³⁰ Assael, M. J. and Polimatidou, S. K. Measurements of the viscosity of alcohols in the temperature range 290–340 K at pressures up to 30 MPa. *Int. J. Thermophys.* **15**(1), 95–107 (1994).
- ³¹ Eicher, L. D. and Zwolinski, B. J. Molecular structure and shear viscosity. isomeric hexanes. *J. Phys. Chem.* **76**(22), 3295–3300 (1972).
- ³² Clark, W. M. and Rowley, R. L. The mutual diffusion coefficient of methanol-n-hexane near the consolute point. *AIChE Journal* **32**(7), 1125–1131 (1986).
- ³³ Maruyama, K., Kawase, S., Tamaki, S., and Okazaki, H. Thermodynamic aspects of the Rayleigh and Brillouin scattering from a binary liquid mixture: The hexane-methanol system. *J. Phys. Chem.* **99**(26), 10644–10647 (1995).